

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Civil Engineering

2. **Subject Name:** Geotechnical Engineering Laboratory L-T-P: 0-0-3
Credits: 2

3. **Pre-requisites:**

4. **Syllabus and reference books:**

Syllabus:

Specific gravity and in-situ density tests, Particle size analyses by sieve and hydrometer, Atterberg's limits tests, Constant and falling head permeability tests, Proctor compaction test, Direct shear test, Unconfined Compressive Strength test and vane shear tests, Unconsolidated Undrained Triaxial compression test, Consolidation test, Demonstration of advanced testing methods

Reference Books:

- 1) Lambe, T. W. 1951. Soil Testing for Engineers. John Wiley & Sons, Inc., New York.
- 2) Bowles, J. E., 1988. Engineering Properties of Soil and Their Measurements. McGraw-hill.
- 3) Indian standards and International Codes of practice for soil testing.

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Specific gravity and in-situ density tests	3
2.	Particle size analyses by sieve and hydrometer	3
3.	Atterberg's limits tests	3
4.	Constant and falling head permeability tests	3
5.	Proctor compaction test	3
6.	Direct shear test	3
7.	Unconfined Compressive Strength test and vane shear tests	3
8.	Unconsolidated Undrained Triaxial compression test	3
9.	Consolidation test	3
10.	Demonstration of advanced testing methods	3
Total number of hours		30